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Inventor(s)	Hussell; Christopher P.

Light-emitting diode packages with transformation and shifting of pulse width modulation signals and related methods

Abstract

Light-emitting diode (LED) packages and, more particularly, LED packages with transformation and shifting of pulse width modulation (PWM) signals and related methods are disclosed. Discrete LED packages are arranged for cascade communication. Each LED package includes one or more LED chips, and each LED package is separately capable of receiving communication from a data stream, controlling operation of the one or more LED chips, and performing transformation and shifting of received PWM signals. Transformation may include compression and/or decompression of received PWM signals by each LED package to provide increased accessible dynamic range. After transformation, LED packages are capable of shifting transformed values to compensate for turn-on delay of LED chips, thereby reducing light drop-off problems at low intensity levels.

Inventors:	Hussell; Christopher P. (Raleigh, NC)
Applicant:	CreeLED, Inc. (Durham, NC)
Family ID:	90829282
Assignee:	CreeLED, Inc. (Durham, NC)
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Primary Examiner: Adams; Carl

Attorney, Agent or Firm: Withrow & Terranova, P.L.L.C.

Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/298,049, filed Apr. 10, 2023, now U.S. Pat. No. 12,014,677, the disclosure of which is hereby incorporated herein by reference in its entirety.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates to light-emitting diode (LED) packages and, more particularly, to LED packages with transformation and shifting of pulse width modulation signals and related methods.

BACKGROUND

(2) Light-emitting diodes (LEDs) are solid-state devices that convert electrical energy to light and generally include one or more active layers of semiconductor material (or an active region) arranged between oppositely doped n-type and p-type layers. When a bias is applied across the doped layers, holes and electrons are injected into the one or more active layers where they recombine to generate emissions such as visible light or ultraviolet emissions.

(3) LEDs have been widely adopted in various illumination contexts, for backlighting of liquid crystal display (LCD) systems (e.g., as a substitute for cold cathode fluorescent lamps) and for direct-view LED displays. Applications utilizing LED arrays include vehicular headlamps, roadway illumination, light fixtures, and various indoor, outdoor, and specialty contexts. Desirable characteristics of LED devices include high luminous efficacy and long lifetime.

(4) Large format multi-color direct-view LED displays (including full color LED video screens) typically include numerous individual LED panels, packages, and/or components providing image resolution determined by the distance between adjacent pixels or “pixel pitch.” Direct-view LED displays typically include three-color displays with arrayed red, green, and blue (RGB) LEDs, and two-color displays with arrayed red and green (RG) LEDs. Other colors and combinations of colors may be used. For many LED display systems, it is desirable to form LED color groups for each pixel such as primary colors red, green, and blue (RGB) that define vertices of a triangle (or polygon) on a chromaticity diagram. This polygon defines the so-called color gamut of the display device, the area of which describes all the possible colors that the display device is capable of producing. Driver printed circuit boards for controlling LED displays are typically densely populated with electrical devices including capacitors, field effect transistors (FETs), decoders, microcontrollers, and the like for driving the pixels of the display. As pixel pitches continue to decrease for higher resolution displays, the density of such electrical devices scales higher corresponding to the increased number of pixels for a given panel area. This tends to add higher complexity and costs to LED panels for display applications.

(5) The art continues to seek improved LED array devices with small pixel pitches while overcoming limitations associated with conventional devices and production methods.

SUMMARY

(6) The present disclosure relates to light-emitting diode (LED) packages and, more particularly, to LED packages with transformation and shifting of pulse width modulation (PWM) signals and related methods. Discrete LED packages are arranged for cascade communication. Each LED package includes one or more LED chips, and each LED package is separately capable of receiving communication from a data stream, controlling operation of the one or more LED chips, and performing transformation and shifting of received data, thereby producing modified PWM signals. Transformation may include compression and/or decompression of received data by each LED package to provide increased accessible dynamic range. After transformation, LED packages are capable of shifting transformed values to compensate for turn-on delay of the LED chips, thereby reducing light drop-off problems at low intensity levels.

(7) In one aspect, a method of light output control within an LED package comprises: transforming a received input brightness value to provide a transformation result; applying a shift value to the transformation result to provide a pulse width modulation (PWM) value; and producing a PWM signal with a pulse width according to the PWM value. The method may further comprise electrically activating an LED chip within the LED package with the PWM signal, wherein the shift value is configured such that the pulse width of the PWM signal is greater than a turn-on time for the LED chip for all possible PWM values. In certain embodiments, transforming the received input brightness value comprises thermal compensation. In certain embodiments, transforming the received input brightness value comprises data decompression. In certain embodiments, the data decompression models a gamma function. In certain embodiments, transforming the received input brightness value comprises multiplication by a calibration factor. In certain embodiments,

producing the PWM signal with the pulse width according to the PWM value comprise a comparison with a PWM counter value, wherein the PWM counter value is updated at a rate according to a clock frequency. In certain embodiments, the clock frequency is determined by a local clock of the LED package. In certain embodiments, the clock frequency is determined by a communication speed of a digital signal received by the LED package. In certain embodiments, the clock frequency is determined by an external clock. In certain embodiments, applying the shift value to the transformation result comprises only applying the shift value when the transformation result is above zero. In certain embodiments, the shift value is applied in a manner so that the PWM value saturates and does not exceed a maximum value of a PWM period and does not roll over to low values.

(8) In another aspect, an LED package configured for receiving and transmitting digital communication signals comprises: at least one LED chip; an LED driver configured to at least partially drive the at least one LED chip by pulse width modulation (PWM); and a PWM processor configured to provide a transformation result based on at least one of multiplication or exponential transformation of a received input brightness value, the PWM processor further configured to provide a PWM value based on a shift of the transformation result. In certain embodiments, the PWM processor is configured to produce a PWM signal to the LED driver based on the PWM value. In certain embodiments, a portion of the PWM signal leaving the PWM processor is zero when the transformation result is zero. In certain embodiments, the PWM processor is configured to provide the PWM value as a final operation before conveyance of the PWM signal to the LED driver. In certain embodiments, the PWM processor comprises a transformation element configured to provide the transformation result and a shifter element configured to provide the PWM value. In certain embodiments, the PWM processor comprises a processing element that is configured to provide both the transformation result and the PWM value. In certain embodiments, the processing element is configured to perform multiple calculations at different times. In certain embodiments, the processing element is configured to perform the multiple calculations under the control of a finite state machine. In certain embodiments, the processing element forms at least a portion of an arithmetic logic unit (ALU). In certain embodiments, the PWM processor is configured such that the PWM value does not exceed a maximum value of a PWM period and the PWM value does not roll over to a lower value. In certain embodiments: the PWM processor is configured to provide a PWM signal to the LED driver with a pulse width according to the PWM value; and the pulse width of the PWM signal is greater than a turn-on time for the at least one LED chip for all levels of a PWM period of the PWM signal.

(9) In another aspect, any of the foregoing aspects individually or together, and/or various separate aspects and features as described herein, may be combined for additional advantage. Any of the various features and elements as disclosed herein may be combined with one or more other disclosed features and elements unless indicated to the contrary herein.

(10) Those skilled in the art will appreciate the scope of the present disclosure and realize additional aspects thereof after reading the following detailed description of the preferred embodiments in association with the accompanying drawing figures.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

(1) The accompanying drawing figures incorporated in and forming a part of this specification illustrate several aspects of the disclosure, and together with the description serve to explain the principles of the disclosure.

(2) FIG. 1 is a block diagram illustrating a system level control scheme for a lighting device using cascade communication for serially connected light-emitting diode (LED) packages

- (3) FIG. 2A is a plot representing several pulse widths corresponding to low-level brightness input values of a pulse width modulation (PWM) signal.
- (4) FIG. 2B is a plot representing light output of LEDs corresponding to the various low-level brightness values and associated pulse widths of FIG. 2A.
- (5) FIG. 3A is a plot illustrating a comparison with and without brightness output loss with a linear brightness scale for digital values of an exemplary 16-bit input grey scale.
- (6) FIG. 3B is a plot of the same data as in FIG. 3A, just with a logarithmic brightness scale to better illustrate a human eye response.
- (7) FIG. 3C is a plot of the brightness intensity versus the PWM counts to illustrate lower level brightness roll-off attributed to delayed LED turn-on times.
- (8) FIG. 4 is a block diagram of an LED package from FIG. 1 capable of transforming and shifting input PWM signals according to principles of the present disclosure.
- (9) FIG. 5 is block diagram of a PWM processor that may represent any one or all of the PWM processors of FIG. 4.

DETAILED DESCRIPTION

- (10) The embodiments set forth below represent the necessary information to enable those skilled in the art to practice the embodiments and illustrate the best mode of practicing the embodiments. Upon reading the following description in light of the accompanying drawing figures, those skilled in the art will understand the concepts of the disclosure and will recognize applications of these concepts not particularly addressed herein. It should be understood that these concepts and applications fall within the scope of the disclosure and the accompanying claims.
- (11) It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present disclosure. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.
- (12) It will be understood that when an element such as a layer, region, or substrate is referred to as being “on” or extending “onto” another element, it can be directly on or extend directly onto the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” or extending “directly onto” another element, there are no intervening elements present. Likewise, it will be understood that when an element such as a layer, region, or substrate is referred to as being “over” or extending “over” another element, it can be directly over or extend directly over the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly over” or extending “directly over” another element, there are no intervening elements present. It will also be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.
- (13) Relative terms such as “below” or “above” or “upper” or “lower” or “horizontal” or “vertical” may be used herein to describe a relationship of one element, layer, or region to another element, layer, or region as illustrated in the Figures. It will be understood that these terms and those discussed above are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures.
- (14) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes,” and/or “including” when used herein specify the presence of stated features, integers, steps,

operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(15) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms used herein should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(16) Embodiments are described herein with reference to schematic illustrations of embodiments of the disclosure. As such, the actual dimensions of the layers and elements can be different, and variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are expected. For example, a region illustrated or described as square or rectangular can have rounded or curved features, and regions shown as straight lines may have some irregularity. Thus, the regions illustrated in the figures are schematic and their shapes are not intended to illustrate the precise shape of a region of a device and are not intended to limit the scope of the disclosure. Additionally, sizes of structures or regions may be exaggerated relative to other structures or regions for illustrative purposes and, thus, are provided to illustrate the general structures of the present subject matter and may or may not be drawn to scale. Common elements between figures may be shown herein with common element numbers and may not be subsequently re-described.

(17) The present disclosure relates to light-emitting diode (LED) packages and, more particularly, to LED packages with transformation and shifting of data used to produce pulse width modulation (PWM) signals and related methods. Discrete LED packages are arranged for cascade communication. Each LED package includes one or more LED chips, and each LED package is separately capable of receiving communication from a data stream, controlling operation of the one or more LED chips, and performing transformation and shifting of received PWM signals. Transformation may include compression and/or decompression of received data by each LED package to provide increased accessible dynamic range. After transformation, LED packages are capable of shifting transformed values to compensate for turn-on delay of the LED chips, thereby reducing light drop-off problems at low intensity levels.

(18) In cascade digital communication, multiple electronic devices are arranged as repeaters to successively receive serial communication for operation. In the context of fine-pitch video displays, multiple LED packages are serially arranged as LED pixels to receive cascade communication. Incoming signals to each LED pixel are produced by another element, such as a master controller or the previous LED pixel, and the bitstream of incoming signals is derived from clock domains of one or more preceding devices. Proper distribution of communication signals to thousands of LED pixels creates challenges. Small sizes are required for LED packages to form pixels of high-resolution video displays and these size constraints provide further challenges.

(19) As used herein, the terms “data stream” and “communication channel” may at times be used interchangeably. However, a “data stream” generally refers to a non-physical representation of data over time that flows through a set of at least one communication channel as well as the internal wiring and storage registers within various elements such as controllers and active electrical elements. A data stream may also be referred to as digital communication between two elements, such as a controller element that transmits digital communication and a receiver element that receives the digital communication. A “communication channel” generally refers to a physical medium through which the data stream is conveyed. For example, a communication channel may comprise a wire with associated electrical elements, an optical fiber, or even air as in the case of radio, light, or sound waves. A given physical channel could also be divided up in time or frequencies to allow multiple “communication channels” within one medium at once such as changing to a different frequency band. In certain aspects, communication channels may embody

serial digital communication channels. Certain aspects relate to a binary communication channel that is a single wire referenced to a common conductor such as ground, which commonly can only hold one value at a time which is high or low voltage (e.g., digital “0” or “1”) and is controlled by the output register of the preceding device. Two-wire differential signaling methods are also contemplated, but the preferred embodiment shown here refers to the single-wire approach primarily because of the added complexity of providing more traces with fine pitch displays.

(20) In certain aspects, the present disclosure relates to light-emitting devices including LEDs, LED packages, and related LED displays and, more particularly, to active control of LEDs within LED displays. LED displays may include rows and columns of LEDs that form an array of LED pixels. A particular LED pixel may include a cluster of LED chips of the same color or multiple colors, with an exemplary LED pixel including a red LED chip, a green LED chip, and a blue LED chip. In certain embodiments, an LED package includes a plurality of LED chips that form at least one LED pixel, and a plurality of such LED packages may be arranged to form an array of LED pixels for an LED display. Each LED package may include its own active electrical element that is configured to receive a control signal and actively maintain an operating state, such as brightness or grey level or a color select signal for the LED chips of the LED device while other LED devices are being addressed. In certain embodiments, the active electrical element may include active circuitry that includes one or more of a driver device, a signal conditioning or transformation device, a memory device, a decoder device, an electrostatic discharge (ESD) protection device, a thermal management device, and a detection device, among others. The active electrical element further includes circuitry to facilitate communication with multiple uncorrelated clock domains, including an original clock domain from a controller and a local clock domain derived within the active electrical element. In this regard, each LED pixel of an LED display may be configured for operation with active matrix addressing with mixed clock domain communication. The active electrical element may be configured to receive one or more of an analog control signal, an encoded analog control signal, a digital control signal, and an encoded digital control signal. In such arrangements, strings of LED packages, each with their own active electrical element, may be arranged for serial communication where each active electrical element receives data from a data stream and transmits data to the next active electrical element in the string of LED packages.

(21) For active matrix addressing, each LED pixel is configured to actively maintain an operating state or otherwise control the driving state, such as brightness or grey level or color select, while other LED pixels are being addressed, thereby allowing each LED pixel to maintain or otherwise independently control their driving state and provide improved display and/or image recording with photographic equipment by reducing or eliminating effects caused by lower-frequency pulsing beating of the display light output with other non-synchronized equipment (e.g., lighting sources, other pulsed displays, or image capture equipment). Accordingly, each LED pixel may be configured to hold its respective operating state with a continuous drive signal, inclusive of pulse-width modulation (PWM), rather than by conventional methods using time division multiplexed signals scanning among groups of pixels that often result in the addition of low frequency components to the drive signals associated with passive matrix addressing. In this regard, each LED pixel may include an active electrical chip or an active electrical element that may include a memory device and the ability to alter a driving condition of the LED pixel based on a state stored in the memory of the active electrical element. In certain embodiments, the continuous drive signal is a constant analog drive current, and in other embodiments where the brightness level may be controlled by pulsed methods such as PWM, the continuous drive signal may refer to a PWM signal that is not interrupted by the time division multiplexed scanning of other LED pixels within the array or within a sub-array. In certain embodiments, the active electrical element may include active circuitry that includes one or more of a driver device, a signal conditioning or transformation device, a memory device, a decoder device, an ESD protection device, a thermal management device, a detection device, and a voltage and/or current sensing device, a command processing

device, and circuitry, among others. In various embodiments, an active electrical element comprises an integrated circuit chip, an application-specific integrated circuit (ASIC), a microcontroller, or a field-programmable gate array (FPGA). In certain embodiments, active electrical elements may be configured to be programmable or reprogrammable after they are manufactured through various memory elements and logic that are incorporated within the active electrical elements.

(22) As used herein, the terms “active electrical chip,” “active electrical element,” or “active electrical component” include any chip or component that is able to alter a driving condition of an LED based on memory or other information that may be stored within a chip or component. As used herein, the terms “active LED pixel” and “smart LED pixel” may be used interchangeably and may refer to a device that includes one or more LED devices or chips that form a pixel and an active electrical element or chip as described above. In certain embodiments, each LED pixel may comprise a single LED package that is configured as an active LED package that includes multiple LED chips and an active electrical element as described above. In this manner, the number of separate electrical devices needed for the LED display may be reduced, such as the separate electrical devices located on the backsides of LED panels of the LED display as previously described. Additionally, overall operating powers needed for operation of the LED panels may be reduced.

(23) Performance of LED displays continues to advance. Previously, LED displays were more adept for static images such as LED signs rather than dynamic application such as video displays since many of the performance metrics expected for good video image display were lacking. Various performance metrics needed for consideration of LEDs for high-quality video display include resolution, contrast, viewing angle, dynamic range, brightness, frame rate, and color gamut, among others. Recent advancements in LED packages, improvements in LED driver quality, and cost reductions have greatly improved the resolution, among other requirements. As the resolution has increased, the packing density of LED packages, such as on printed circuit boards (PCBs) has also increased and become more complex. The driving of LEDs within LED packages to achieve high dynamic range simultaneously with high frame and refresh rates remains challenging. This is because current driving techniques require remote drivers placed on the opposing side of PCBs containing LED arrays. Packing density constraints dictate that the drivers need to be shared via time division multiplexing techniques, such as raster scan. As the preferred driving techniques for LEDs utilize PWM to set the brightness, ever-higher frequencies are required to achieve high dynamic range. Consequently, the dynamic range is limited by the highest frequency pulse that can be delivered through in view of with parasitic resistance, capacitance, and inductance. LED packages arranged as pixels for cascade serial communication afford the opportunity to provide drivers local to each LED package, thereby providing improved dynamic range by removing the need for shared drivers and greatly reducing the parasitic resistance, capacitance, and inductance.

(24) FIG. 1 is a block diagram **10** illustrating a system level control scheme for a lighting device using cascade communication for serially connected LED packages **12**. The lighting device may embody an LED display and each LED package **12** may form an LED pixel of the display. For such applications, the terms LED package and LED pixel may be used interchangeably, although it is recognized that an LED package may be composed of several LED pixels formed together in one component. An exemplary LED string **14** arranged for serial communication is indicated by a dashed box in FIG. 1. While only the single LED string **14** is provided in detail, one or more other LED strings may also be coupled with a controller **16**. As illustrated, the controller **16** is arranged to control one or more LED strings **14**. The controller **16** may comprise an integrated circuit, such as one or more of an ASIC, a microcontroller, a programmable control element, and an FPGA. In certain embodiments, the controller **16** may be referred to as a master controller for the LED string **14**. In other embodiments, the controller **16** may be a sub-controller to which another master controller (not shown) delegates a set of tasks as it pertains to a larger system. A data signal out (DOUT) of the controller **16** may be passed along the LED string **14** in a serial manner, and a

return data signal in (DIN) may be received back by the controller **16**. The signal includes an original clock domain provided by the controller **16** or another master controller as described above. In FIG. **1**, each LED package **12**, or LED pixel, is provided with a label such as “Px 1,1” where the first number represents a row, and the second number represents a column. Each LED package **12** includes its own active electrical element **18** that is registered and housed therewithin so that each LED package **12** comprises logic for responding to received data signals.

(25) According to the arrangement of FIG. **1**, important aspects include delivery of high bit depth data to incorporated LED drivers within each LED package **12** and being capable of converting data effectively to energize the LED chips within each LED package **12** according to expected light output levels with increased dynamic range. In certain aspects, LED packages **12** and associated active electrical elements **18** are capable of transforming received data by way of compression and/or decompression techniques such as gamma function or correction. However, as this transformed data is used for input to a PWM driver, lower values may not produce the expected response from LED chips due to pulse width being shorter than time delay required to energize the LED chips. This can happen over several of the lowest brightness levels resulting in a dramatic reduction of the dynamic range. As disclosed herein, LED packages **12** and related methods are disclosed where integrated LED drivers are controlled such that the dynamic range from bit depths is better achieved as expected. As described later in greater detail, LED packages **12** are disclosed that are capable of transforming received data and provide increased dynamic range by shifting transformed values sufficiently to compensate for time delay required to energize the LED chips. By applying shift values to transformed data, pulse widths that are shorter than turn-on time delay for LED chips may be avoided. As such, with proper selection of shift values, all data values conveyed to the PWM driver produce light output from the respective LED chips.

(26) FIG. **2A** is a plot representing several pulse widths corresponding to low-level brightness input values of a PWM signal. The y-axis represents a PWM signal that may be provided to the current or power drivers internal to the LED packages **12** of FIG. **1**. In certain aspects, the PWM signal may be the output of a PWM processor as later described herein. The x-axis represents relative time to show a comparison of pulse widths. Pulse widths for brightness values or levels of 7, 10, 14, and 20 are illustrated that represent lower brightness values for any type of bit depth, such as 8-bit or 16-bit, among others. For example, the values of 7, 10, 14, and 20 could represent lower brightness levels out of 255 levels for 8-bit signals or out of 65,535 levels for 16-bit levels. As illustrated, the pulse width increases with increasing brightness level.

(27) FIG. **2B** is a plot representing light output of LEDs corresponding to the various low-level brightness values and associated pulse widths of FIG. **2A**. The y-axis represents the light output and the x-axis represents relative time on a same scale as FIG. **2A**. For each of the values 7, 10, 14, and 20, a same delay is realized before the LEDs turn-on and produce light output. Contributing factors to this delay mainly include an amount of resistance and capacitance in the driving circuit of the LED package and a forward voltage of the LED chip above which the LED chip begins emitting light. This forward voltage required to produce light depends on the bandgap of the active region of the LED chip and accordingly may also correspond to the color of light emitted. This no-light-output delay time is the time that it takes for the driver to charge the various capacitances of the driver output, control lines, etc., as well as the capacitance of the LED itself, thereby bringing the voltage from a lower level up to the forward voltage required to produce light. This delay time may be on the order of 1 microsecond (μ s) or even much less, but PWM counter frequencies required to produce high dynamic range are much shorter. For example, PWM counter or clock frequencies on the order of 10 megahertz (MHz) or greater may be required.

(28) When comparing the pulse widths of FIG. **2A** with the light output responses of FIG. **2B**, problems of little or no light output for low-level brightness values are apparent. As illustrated in FIG. **2B**, the delay time may be considerable enough such that values of 7 or below produce very little or no LED light output before the end of the pulse width of FIG. **2A**. Values between 8 and 14

produce successively increasing amounts of light before the end of their respective pulse widths, and values above 14 generally produce constant increments in the amount of light output since the LEDs are fully turned on for larger portions of the pulse widths. At the ends of the respective pulse widths of FIG. 2A, the corresponding light outputs of FIG. 2B decrease rapidly, but not instantly, due to discharge of capacitance as the LEDs turn off. At turn-off, voltages will drop near bandgaps of LED chips and certain capacitances may hold voltage levels above zero such that next PWM pulses may exhibit shorter delay than at initial start-up. Despite such shorter delay, the problems of little or no light output for low-level brightness values are noticeable considering the integration of many pulses and many associated delays during PWM control.

(29) FIGS. 3A and 3B are plots illustrating a comparison with and without brightness output loss for digital values of an exemplary 16-bit input grey scale used by an exemplary PWM driver. Such brightness output loss is described above in the context of FIGS. 2A and 2B where for illustrative purposes the first 16 values are abruptly lost with zero output. For both FIGS. 3A and 3B, the x-axis represents brightness input values and the y-axis represents brightness output in arbitrary units. In FIG. 3A, the y-axis is provided as a linear axis while in FIG. 3B, the same data is provided with a logarithmic y-axis. The 16-bit input, or 65,535 levels, yields a dynamic range of nearly five orders of magnitude. A solid line is provided for brightness output loss of the first 16 levels in comparison with a dashed line without brightness output loss. In FIG. 3A, the data lines without brightness output loss and with 16-level loss appear to overlap. In this regard, if the human eye response were linear, such brightness loss may not be noticeable. However, the human eye response is better approximated by logarithmic behavior. As such, FIG. 3B illustrates how LED delay times that cause brightness loss for just the first 16 levels adversely correlate to the loss of more than one order of magnitude of the overall dynamic range.

(30) FIG. 3C is a plot of input PWM counts versus brightness intensity in arbitrary units to illustrate lower level brightness roll-off attributed to delayed LED turn-on times. In the example of FIG. 3C, roll-off from intended brightness begins below a PWM count of about 10 and the PWM value of “1” is completely lost. In this manner, the pulse width associated with the PWM value of “1” is smaller than the time delay for turning on the LED. In applications where the PWM frequency is increased to obtain more dynamic range and/or increased refresh rate, the intensity roll-off or dropout would be more severe and shifted to the right to encompass even more PWM counts.

(31) According to aspects disclosed herein, the aforementioned problems associated with delayed LED turn-on may be compensated by shifting transformed input data so that each output PWM value corresponds with a pulse width that is at least greater than a delay time for the LED chip. In this manner, suitable brightness output may be provided across the dynamic range, including at the lower levels described above. LED packages arranged as LED pixels in a display may each include the capability to transform an input PWM signal for increased dynamic range and then shift the transformed values to avoid smaller pulse widths associated with brightness loss at lower levels.

(32) FIG. 4 is a block diagram of an LED package 12 from FIG. 1 capable of transforming and shifting input PWM signals according to principles of the present disclosure. The active electrical element 18 may include multiple ports represented by a supply voltage (Vdd), ground (GND or Vss), and bidirectional communication ports or digital input/output ports (DIO1 and DIO2) according to embodiments disclosed herein. By having the DIO1 and DIO2 ports configured as bidirectional ports, the active electrical element 18 may advantageously be able to detect an input signal from a communication channel and then assign one of the DIO1 and DIO2 ports as an input port and the other of the DIO1 and DIO2 ports as the output port. Such functionality may be provided by input/output buffers and/or an active switching network internal to the active electrical element and electrically coupled to the DIO1 and DIO2 ports. This provides flexibility in layouts for displays where a plurality of LED packages 12 are connected together for cascade communication. For example, multiple LED packages 12 may be arranged in multiple rows where

data cascades from package-to-package along each row and in a serpentine manner from row-to-row as illustrated in FIG. 1. In such arrangements, the bidirectional communication ports allow the LED packages 12 to be mounted in a same orientation and receive and transmit digital communication left-to-right or right-to-left depending on the row position. In addition to the four ports of Vdd, GND, DIO1, and DIO2 on the left side of the block diagram, the active electrical element 18 includes four ports on the right side that are coupled with LEDs 20-1 to 20-3 of the LED package 12. In this regard, the LEDs 20-1 to 20-3 are packaged together with the active electrical element 18 in the common LED package 12 to form an individual pixel of a larger display. As used herein, the LEDs 20-1 to 20-3 may also be referred to as LED chips.

(33) Certain elements of the active electrical element 18 are described below; however, it is understood that the active electrical element 18 may include many other components, including memory elements, signal conditioning elements, thermal management, electrostatic discharge elements, clock elements, and oscillators, among others. In FIG. 4, control logic 22 is arranged to receive input data, execute commands according to a command protocol, provide control signals for operation of the LEDs 20-1 to 20-3, report various voltage levels and/or temperature levels included with output data, and transmit the output data via the DIO1 and DIO2 ports to the next adjacent LED package. The control logic 22 may operate in the digital domain and may include input/output buffers electrically coupled to the DIO1 and DIO2 ports that assign input and output configurations for the bidirectional DIO1 and DIO2 ports.

(34) In certain embodiments, the active electrical element 18 may be configured to provide both forward and reverse bias states to the LEDs 20-1 to 20-3. In this regard, the control logic 22 may include a reverse bias control output signal that, with appropriate active elements, is configured to supply either near-Vdd or near-GND voltage levels to the LEDs 20-1 to 20-3. Since the nomenclature “reverse bias” implies that a high level on the control logic 22 output produces a reverse bias condition, the output signal could simply be coupled with an inverter 24 that is provided in a driver 26 of the active electrical element 18. As such, the LEDs 20-1 to 20-3 may be either forward biased or reverse biased depending on a particular operating state and/or command received by the control logic 22. The inverter 24, or inverter logic element, may have sufficient output characteristics to drive the LEDs 20-1 to 20-3. The driver 26 may be substantially an analog interface of the active electrical element 18 that is electrically coupled with the control logic 22. The driver 26 may include controllable current sources 28-1 to 28-3, which could also be configured as LED sink drivers. Pull-up resistors R1 to R3 may be incorporated to provide paths to Vdd for each of the LEDs 20-1 to 20-3, which aid with the voltage measurement when configured for reverse bias. Each of the current sources 28-1 to 28-3 may be electrically coupled with digital output signals LED1 to LED3 of the control logic 22. The output signals LED1 to LED3 may be provided along multiple wires that are coupled to each of the current sources 28-1 to 28-3 for current selection purposes. The output signals LED1 to LED3 may embody PWM outputs of the control logic 22 for controlling operation of the LEDs 20-1 to 20-3. The driver 26 may also include a multiplexer 30 electrically coupled with an analog-to-digital (ADC) converter and ADC selector of the control logic 22. Additionally, the driver 26 may include an on-chip temperature sensor that is provided through the multiplexer 30. In certain embodiments, the temperature sensor provides thermal compensation for the LEDs 20-1 to 20-3 via a thermal compensation curve and/or thermal shut down.

(35) The active electrical element 18 further comprises a serial interface 32 that embodies a module with circuitry configured to decode and convert the incoming signal of the data stream into a bitstream in a local clock domain, which can be further processed by the control logic 22. In this manner, the serial interface 32 may also be referred to as a digital communication receiving device. The digital communication could be received from a controller (e.g., 16 of FIG. 1) And/or another LED package (e.g., 12 of FIG. 1) in a serial string. The serial interface 32 is further configured to retransmit the decoded and converted bitstream along with modified data to the communication

channel to which another LED package or another external element is connected in a manner that is compatible with the overall LED display system.

(36) In certain embodiments, the control logic **22** may include circuitry in the form of one or more PWM processors **34-1** to **34-3** that provide the output PWM signals LED**1** to LED**3**. A separate PWM processor **34-1** to **34-3** may be provided for each LED chip **20-1** to **20-3** or the PWM processors **34-1** to **34-3** may be combined as single PWM processor for all of the LED chips **20-1** to **20-3**. As will be described below in greater detail for FIG. 5, the one or more PWM processors **34-1** to **34-3** are configured to transform input PWM signals and then shift the transformed PWM values so pulse widths are provided to the LED chips **20-1** to **20-3** that compensate for delayed turn-on times.

(37) FIG. 5 is block diagram of a PWM processor **34** that may represent any one or all of the PWM processors **34-1** to **34-3** of FIG. 4. Hash marks across various conductor lines indicate that multiple lines or signals may also be provided. The PWM processor **34** is configured to receive a brightness input value, such as input data, along with additional inputs labeled Other Settings and Controls, Shift Value, and PWM counter from other elements of the control logic **22** of FIG. 4. The PWM processor **34** outputs the PWM signal as a train of pulses for other elements, particularly the driver **26** of FIG. 4. The PWM counter input may embody a PWM counter value that is updated at a rate according to a clock frequency. In certain embodiments, the clock frequency is determined by a local oscillator of a clock of the active electrical element **18** of FIG. 4. The clock frequency may also be determined by a communication speed of the received digital signal. In still further embodiments, the clock frequency is determined by a clock that is external to the LED package **12** of FIG. 4, such as the controller **16** of FIG. 1.

(38) In FIG. 5, a transform element **36** may perform calculations according to the inputs. Exemplary calculations include compression or decompression to provide increased dynamic range. Transformation results may include at least one of multiplication or exponential transformation of a received input brightness value. In one aspect, the calculations are gamma calculations. In certain embodiments, the transform element **36** may provide a transform output, or transformation result, according to the formula

(39) $\text{transformoutput} = \text{CalFctr} * f(\text{brightness}) * g(\text{scaledTemp})$

(40) where CalFctr is the calibration factor, f is a transfer function providing capability such as data decompression (e.g., gamma) to the brightness input value, g is another transfer function providing temperature compensation capability in conjunction with the scaledTemp value received by a temperature sensor. The scaledTemp value may be determined from raw temperature sensor values according to the formula

(41) $\text{scaledTemp} = \text{TempScale} * (\text{rawTemp} - \text{TempShift})$

(42) where rawTemp is the raw data value coming from the temperature sensor.

(43) As illustrated in FIG. 5, the transform output of the transform element **36** is received by a shifter element **38**. Based on the Shift Value input received, the shifter element **38** is configured to shift the transform output in a manner that compensates for LED turn-on delay. The Shift Value input may be a value that is hard set into the active electrical element **18** of FIG. 4, or the Shift Value may be user selectable. In certain embodiments, the Shift Value may be determined by an initial calibration that identifies a lowest PWM level that exhibits a pulse width that still provides useful light output. Alternatively, the Shift Value may be the result of previous calibrations for similar systems and/or based on known electrical characteristics of LED chips selected. Useful light output depends on application, but generally refers to a light output at a level that would be noticeable to a user. In another manner, a lowest PWM level that still provides useful light may be characterized where the corresponding pulse width is greater than a time delay to turn on the LED chip. Once the lowest useful PWM level is determined, the Shift Value may then be set at the lowest useful PWM level or even above it to ensure light output across the entire dynamic range. The Shift Value may then be user selectable through a communication system to the PWM

processor **34**, or hard set or coded based on system design. In certain embodiments, the shifter element **38** may comprise a logic set of gates, such as AND gates, OR gates, or other logic. The shifter element **38** may embody a processor element executing a program that resides within the PWM processor **34**. In certain aspects, logic associated with the shifter element **38** may be shared with other elements of the active electrical element **18**. In certain aspects, the shifter element **38** may also be referred to as an adder.

(44) As illustrated in FIG. 5, the PWM processor **34** is configured to apply the shift after the input data (e.g., Brightness Input Value) are transformed by the transform element **36**. The transformed and shifted signal is then provided as a PWM value **40** that is a result of at least two processing elements (i.e., the transform element **36** and the shifter element **38**). The output PWM signal of the PWM processor **34** is then generated as the result of a digital comparator **42** which outputs an “LED on” signal when the PWM counter input is less than the PWM value **40** and “LED off” otherwise. Accordingly, the PWM value **40** may be referred to as a post-shifted value before the output PWM signal is provided to the driver **26** of FIG. 4.

(45) Without the shift provided by the shifter element **38**, the transform output from the transform element **36** would be provided to the digital comparator **42** for comparison with the PWM counter. Since the transform output is intended to increase resolution at lower PWM levels, such an arrangement would output various PWM levels below LED turn-on thresholds. Alternatively, if a shift was implemented before transformation in the transform element **36**, such as a shift already being present in the input brightness value received by the controller **16** of FIG. 1, multiplication factors during transformation may still result in PWM levels with pulse widths below LED turn-on times. In order to provide functionality of calibration, thermal compensation, and/or data decompression while also compensating for LED turn-on times, the shift should be applied following the transformation. This ensures that all transformed values have a pulse width that exceeds the LED turn-on time. In this manner, the PWM value **40** of FIG. 5 may be determined according to the formula

(46) $PWMValue = transformoutput + ShiftValue$

which may also be represented according to the below formula in view of the calculation for the transform output provided above.

(47) $PWMValue = CalFctr * f(brightness) * g(scaledTemp) + ShiftValue$

Accordingly, the PWM value **40** of FIG. 5 may be the result of a first transformation which may include a series of multiplications followed by addition of the Shift Value.

(48) By way of example, for 16-bit signals with levels from 0-65,535, it may be determined that a Shift Value of 10 is needed to compensate for LED turn-on delay. Accordingly, transform output values from the transform element **36** will be shifted 10 levels by the shifter element **38** such that a post-transformation value of “1” is shifted to a value of “11” to ensure useful light output. In certain embodiments, logic within the PWM processor **34** may be implemented so that the post-Shift Value does not exceed a maximum PWM value of a PWM period dictated by the PWM counter. In the 16-bit example above, such logic would shift values from 1 to 65,525 by 10 levels, and reduced shifts would be applied such that all values from 65,525 and up would saturate and equal the maximum of 65,535 levels and not roll over to low values. This may be necessary because low-level logic designs may not accommodate larger values and a high value would roll over to a low value instead. With reference to the logarithmic eye response described in FIG. 3B, differences between levels 65,525 and 65,535 are much less detectable than differences at lower levels between 1 and 10, since a level of 10 appears ten times brighter than a level of 1. In still further embodiments, the logic within the PWM processor **34** may also be implemented to suppress shifting for transform output values of “0” to ensure there is no light output as intended for values of zero. With implementation of such logic, the PWM value **40** of FIG. 5 may be determined according to the formula

$$(49) \text{ PWMValue} = \text{transformoutput} + \begin{cases} \text{PWM}_{\text{Max}} & \text{iftransformoutput} + \text{ShiftValue} > \text{PWM}_{\text{Max}} \\ \text{ShiftValue} & \text{otherwise} \end{cases}$$

such that values of “0” remain “0” and PWM values **40** do not exceed maximum PWM values as dictated by the PWM counter. In this regard, shifter element **38** may be configured as an adder set to operate only if transform output > 0 and produces PWM.sub.Max when a final carry signal is detected.

(50) The description provided above may convey various aspects of the present disclosure according to various embodiments. Many other adaptations are contemplated. For example, individual PWM processors **34** may be provided for each LED chip within a package, or the function of individual PWM processors **34** may be provided with a single PWM processor **34** providing the various calculations at various times. The transform element **36** and the shifter element **38** may form separate processing elements within the PWM processor **34** or they may be integrated as a single processing element. Such single or multiple processing elements within the PWM processor **34** may be implemented according to the direction of a finite state machine (FSM). A further example would be the single or multiple processing elements forming at least a portion of an arithmetic logic unit (ALU) that provides all the mathematical functions as controlled by an FSM. In view of the above, the block diagram of FIG. 5 may provide one implementation and/or arrangement of physical hardware and connections. Alternatively, the block diagram of FIG. 5 may conceptually provide how physical hardware routes signals over time, but not necessarily how the physical hardware is specifically arranged and physically connected together in various embodiments.

(51) It is contemplated that any of the foregoing aspects, and/or various separate aspects and features as described herein, may be combined for additional advantage. Any of the various embodiments as disclosed herein may be combined with one or more other disclosed embodiments unless indicated to the contrary herein.

(52) Those skilled in the art will recognize improvements and modifications to the preferred embodiments of the present disclosure. All such improvements and modifications are considered within the scope of the concepts disclosed herein and the claims that follow.

Claims

1. A light-emitting diode (LED) display comprising: a display panel; and at least one LED package comprising: at least one LED chip; an LED driver configured to at least partially drive the at least one LED chip by pulse width modulation (PWM); and a PWM processor configured to provide a transformation result based on at least one of multiplication or exponential transformation of a received input brightness value, the PWM processor further configured to provide a PWM value based on a shift of the transformation result, and the shift is based on a lowest PWM signal level that corresponds to a turn-on time for the at least one LED chip, wherein the PWM processor is configured to provide a PWM signal to the LED driver with a pulse width according to the PWM value, and the pulse width of the PWM signal is greater than the turn-on time for the at least one LED chip for all PWM signal levels of a PWM period of the PWM signal.
2. The LED display of claim 1, wherein a portion of the PWM signal leaving the PWM processor is zero when the transformation result is zero.
3. The LED display of claim 1, wherein the PWM processor is configured to provide the PWM value as a final operation before conveyance of the PWM signal to the LED driver.
4. The LED display of claim 1, wherein the PWM processor comprises a transformation element configured to provide the transformation result and a shifter element configured to provide the PWM value.

5. The LED display of claim 1, wherein the PWM processor comprises a processing element that is configured to provide both the transformation result and the PWM value.
 6. The LED display of claim 5, wherein the processing element is configured to perform multiple calculations at different times.
 7. The LED display of claim 6, wherein the processing element is configured to perform the multiple calculations under the control of a finite state machine.
 8. The LED display of claim 6, wherein the processing element forms at least a portion of an arithmetic logic unit (ALU).
 9. The LED display of claim 1, wherein the PWM processor is configured such that the PWM value does not exceed a maximum value of a PWM period and the PWM value does not roll over to a lower value.
 10. A method of light output control for a light-emitting diode (LED) display, the method comprising: sending a data signal from a controller to a plurality of serially connected LED packages; receiving an input brightness value of the data signal within an LED package of the plurality of serially connected LED packages; transforming the input brightness value to provide a transformation result; applying a shift value to the transformation result to provide a pulse width modulation (PWM) value, the shift value being based on a lowest PWM signal level that corresponds to a turn-on time for an LED chip within the LED package; producing a PWM signal with a pulse width according to the PWM value; and electrically activating the LED chip within the LED package with the PWM signal, wherein the shift value is configured such that the pulse width of the PWM signal is greater than the turn-on time for the LED chip for all possible PWM values.
 11. The method of claim 10, wherein transforming the input brightness value comprises thermal compensation.
 12. The method of claim 10, wherein transforming the input brightness value comprises data decompression.
 13. The method of claim 12, wherein the data decompression models a gamma function.
 14. The method of claim 10, wherein transforming the input brightness value comprises multiplication by a calibration factor.
 15. The method of claim 10, wherein producing the PWM signal with the pulse width according to the PWM value comprises a comparison with a PWM counter value, wherein the PWM counter value is updated at a rate according to a clock frequency.
 16. The method of claim 15, wherein the clock frequency is determined by a local clock of the LED package.
 17. The method of claim 15, wherein the clock frequency is determined by a communication speed of a digital signal received by the LED package.
 18. The method of claim 15, wherein the clock frequency is determined by an external clock.
 19. The method of claim 10, wherein applying the shift value to the transformation result comprises only applying the shift value when the transformation result is above zero.
 20. The method of claim 19, wherein the shift value is applied in a manner so that the PWM value saturates and does not exceed a maximum value of a PWM period and does not roll over to a lower value.
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